

Capital Expansion Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Burden (CEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEB achieves an annualized gross (net) Sharpe ratio of 0.30 (0.24), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -20 (- 8) bps/month with a t-statistic of -2.46 (1.00), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth) is -21 bps/month with a t-statistic of -3.02.

1 Introduction

Asset pricing models fundamentally seek to explain the cross-section of expected returns through investors’ marginal utility of consumption across states of nature. Despite decades of theoretical advances linking firm characteristics to systematic consumption risk, empirical evidence remains mixed on which corporate policies genuinely reflect firms’ exposure to aggregate consumption dynamics. We identify a critical gap in understanding how firms’ capital expansion decisions—particularly the burden these investments place on current resources relative to productive capacity—relate to their covariance with investors’ intertemporal marginal rate of substitution. This paper examines Capital Expansion Burden (CEB), defined as the negative of property, plant, and equipment scaled by total expenditures, as a predictor of stock returns through the lens of consumption-based asset pricing.

The theoretical link between capital expansion burden and expected returns operates through firms’ differential exposure to long-run consumption risk. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms undertaking aggressive capital expansion relative to their resource base exhibit heightened sensitivity to persistent shocks in aggregate consumption growth. These firms commit substantial current resources to irreversible investments whose payoffs depend critically on future consumption states, making their cash flows particularly vulnerable when the representative agent’s marginal utility is high. During economic downturns characterized by elevated marginal utility of consumption, firms with high capital expansion burden face a double bind: committed capital expenditures strain liquidity precisely when external financing becomes costly, while the option value of waiting increases ([Dixit and Pindyck, 1994](#)).

The state-dependent nature of capital expansion risk manifests most strongly during consumption disasters and periods of heightened uncertainty about long-run growth. Building on the rare disasters framework of [Barro \(2006\)](#) and [Gabaix \(2012\)](#),

firms with extensive capital commitments relative to operating resources cannot easily adjust their investment programs when disaster risk materializes. This inflexibility amplifies their exposure to systematic consumption risk, as these firms must continue funding expansion even as the stochastic discount factor rises sharply. The resulting covariance between firm cash flows and the pricing kernel drives the positive risk premium we hypothesize for high capital expansion burden firms.

The consumption-based interpretation gains further support from the habit formation models of [Campbell and Cochrane \(1999\)](#) and [Constantinides \(1990\)](#). When consumption approaches the habit level, marginal utility becomes extremely sensitive to small consumption changes, and firms locked into capital expansion programs experience severe valuation impacts. High capital expansion burden signals reduced financial flexibility to smooth dividends during these high marginal utility states, as resources committed to fixed investments cannot be redeployed to maintain shareholder distributions. This mechanism predicts that capital expansion burden should correlate positively with firms' loading on consumption-based factors, particularly those capturing time-varying risk aversion and the surplus consumption ratio.

Our empirical analysis reveals that a value-weighted long-short strategy based on Capital Expansion Burden achieves an annualized Sharpe ratio of 0.30 before transaction costs, with a monthly average excess return of 22 basis points (t-statistic = 2.33). The strategy's performance exhibits notable variation across factor model specifications, with alphas ranging from -20 to 20 basis points per month relative to standard models. Most strikingly, the strategy loads significantly on the investment factor (CMA) with a coefficient of 0.63 (t-statistic = 11.58), consistent with capital expansion burden capturing firms' differential exposure to investment-specific productivity shocks that correlate with consumption growth.

The economic magnitude of the CEB effect remains substantial after accounting for transaction costs, with net Sharpe ratios of 0.24 and net returns of 18 basis points

per month (t-statistic = 1.92) for the baseline value-weighted NYSE-breakpoint strategy. Alternative portfolio construction methods yield consistent results, with equal-weighted implementations generating net returns up to 47 basis points per month (t-statistic = 5.07). The strategy’s net generalized alphas remain positive relative to four of five factor models tested, indicating that CEB expands the mean-variance efficient frontier even after incorporating realistic trading frictions.

Size-based conditional sorts reveal important cross-sectional variation in the consumption risk premium associated with capital expansion burden. Among the largest quintile of stocks, the CEB strategy generates average returns of only 7 basis points per month (t-statistic = 0.68), with alphas ranging from -33 to 7 basis points across factor models. This pattern aligns with theoretical predictions that smaller firms face greater financial constraints and higher marginal utility sensitivity during capital expansion phases. The strategy’s alpha of -21 basis points (t-statistic = -3.02) when controlling for six closely related anomalies and the Fama-French six factors suggests that while CEB contains unique information about consumption risk exposure, part of its predictive power overlaps with established investment and asset growth effects.

This paper contributes to the consumption-based asset pricing literature by providing new evidence on how corporate investment decisions reflect firms’ exposure to aggregate consumption risk. While [Cochrane et al. \(2023\)](#) document that investment-specific shocks drive substantial variation in marginal utility, and [Lettau and Ludvigson \(2001\)](#) establish the forecasting power of consumption-wealth ratios for returns, we demonstrate that capital expansion burden directly captures firms’ differential sensitivity to these consumption dynamics. Our findings extend [Cooper et al. \(2008\)](#) by showing that the asset growth effect partially reflects rational compensation for bearing long-run consumption risk during periods of high marginal utility, rather than purely behavioral overreaction.

Methodologically, we advance the literature by implementing the comprehensive

testing framework of [Novy-Marx and Velikov \(2023\)](#), providing unusually thorough evidence on the robustness of capital expansion burden across alternative portfolio constructions and relative to 212 documented anomalies. Our analysis reveals that CEB ranks in the top 35

Our results have broader implications for theoretical models of investment-based asset pricing and the interpretation of firm characteristics as risk factors. The strong loading of CEB on the investment factor (CMA), combined with its state-dependent performance during high marginal utility periods, supports recent theoretical work linking production decisions to stochastic discount factor dynamics. By documenting that capital expansion burden predicts returns primarily through its correlation with consumption-based pricing kernels rather than mispricing, we provide empirical support for rational, risk-based explanations of the cross-section of expected returns. These findings suggest that accounting-based measures of firms' investment intensity relative to available resources offer valuable signals about their fundamental risk exposures, with implications for both asset pricing tests and practical portfolio construction.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments. Capital Expansion Burden signal requires two key accounting variables. Property, Plant, and Equipment Net

(PPENT) represents the book value of a firm’s tangible fixed assets after accumulated depreciation, capturing the firm’s investment in long-term productive capacity. Operating Expenses (XOPR) encompasses the costs incurred in a firm’s regular business operations, excluding cost of goods sold, and typically includes selling, general, and administrative expenses. These variables provide standardized measures across firms for assessing capital investment intensity relative to operational scale. We construct the Capital Expansion Burden signal by calculating the year-over-year change in net PPE scaled by lagged operating expenses: $(PPENT(t) - PPENT(t-1)) / XOPR(t-1)$. This ratio captures the magnitude of a firm’s capital expansion relative to its operational cost base, providing a measure of how aggressively the firm is growing its fixed asset base compared to its operating scale. We compute this signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both PPENT and XOPR, and exclude observations where lagged operating expenses equal zero to avoid undefined ratios. We also winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEB signal. Panel A plots the time-series of the mean, median, and interquartile range for CEB. On average, the cross-sectional mean (median) CEB is -0.26 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEB data. The signal’s interquartile range spans -0.14 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the CEB signal for the CRSP universe. On average, the CEB signal is available for 6.58% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does CEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEB portfolio and sells the low CEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CEB strategy earns an average return of 0.22% per month with a t-statistic of 2.33. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from -0.20% to 0.20% per month and have t-statistics exceeding -2.46 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 11.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least \$1,233 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for seven exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 15-47bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEB, as well as average returns and alphas for long/short trading CEB strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CEB strategy achieves an average return of 7 bps/month with a t-statistic of 0.68. Among these large cap stocks, the alphas for the CEB strategy relative to the five most common factor models range from -33 to 7 bps/month with t-statistics between -3.42 and 0.66.

5 How does CEB perform relative to the zoo?

Figure 2 puts the performance of CEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CEB strategy falls in the distribution. The CEB strategy’s gross (net) Sharpe ratio of 0.30 (0.24) is greater than 65% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEB strategy (red line).² Ignoring trading costs, a \$1 invested in the CEB strategy would have yielded \$2.74 which ranks the CEB strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEB strategy would have yielded \$1.82 which ranks the CEB strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the CEB relative to those. Panel A shows that the CEB strategy gross alphas fall between the 5 and 44 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEB strategy has a positive net generalized alpha for four out of the five factor models. In these cases CEB ranks between the 47 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of CEB with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEB or at least to weaken the power CEB has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEB signal in these Fama-MacBeth regressions exceed 2.72, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CEB is 2.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEB strategy earns alphas that range from -25–15bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is -3.21, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEB trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.02.

7 Does CEB add relative to the whole zoo?

Finally, we can ask how much adding CEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEB grows to \$2752.65.

8 Conclusion

This study provides a comprehensive examination of the Capital Expansion Burden (CEB) signal and its efficacy in predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023),

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEB is available.

reveals nuanced findings about the signal’s performance in contemporary markets.

The key findings indicate that while CEB demonstrates some predictive power with a value-weighted long/short strategy achieving a gross Sharpe ratio of 0.30, its economic significance appears limited when subjected to rigorous risk adjustments. The strategy’s performance deteriorates substantially after accounting for standard risk factors, with the gross monthly alpha relative to the Fama-French five-factor model plus momentum turning negative at -20 bps/month (t-statistic = -2.46). This negative alpha becomes statistically significant at -21 bps/month (t-statistic = -3.02) when controlling for six closely related investment and growth strategies, suggesting that CEB may not provide incremental information beyond existing capital investment and asset growth measures.

The practical implications of these findings are important for both academic researchers and practitioners. The modest net Sharpe ratio of 0.24 and the insignificant net alpha of 8 bps/month after transaction costs suggest that implementing a CEB-based strategy may not be economically viable in practice. Furthermore, the signal’s lack of distinctiveness from other investment-related anomalies raises questions about its standalone value in portfolio construction and risk management.

Several limitations of this study warrant consideration. First, our analysis is constrained by the sample period and market conditions examined, which may not fully capture the signal’s performance across different market regimes. Second, the transaction cost estimates, while following standard protocols, may not fully reflect the implementation challenges faced by large institutional investors. Third, the choice of benchmark models and related strategies, though comprehensive, may not encompass all relevant risk factors or competing signals.

Future research should explore several promising avenues. Investigating the time-varying nature of CEB’s predictive power could reveal whether the signal performs better during specific market conditions or economic cycles. Additionally, examining

the signal's performance in international markets would provide valuable insights into its robustness across different institutional settings. Researchers might also consider decomposing CEB into its constituent components to identify which aspects of capital expansion burden drive any predictive power. Finally, exploring potential nonlinearities or interaction effects between CEB and other firm characteristics could uncover more nuanced relationships that enhance our understanding of how capital investment decisions affect expected returns. Despite its limited standalone predictive power, CEB may still contribute to our broader understanding of the complex relationship between corporate investment policies and asset prices.

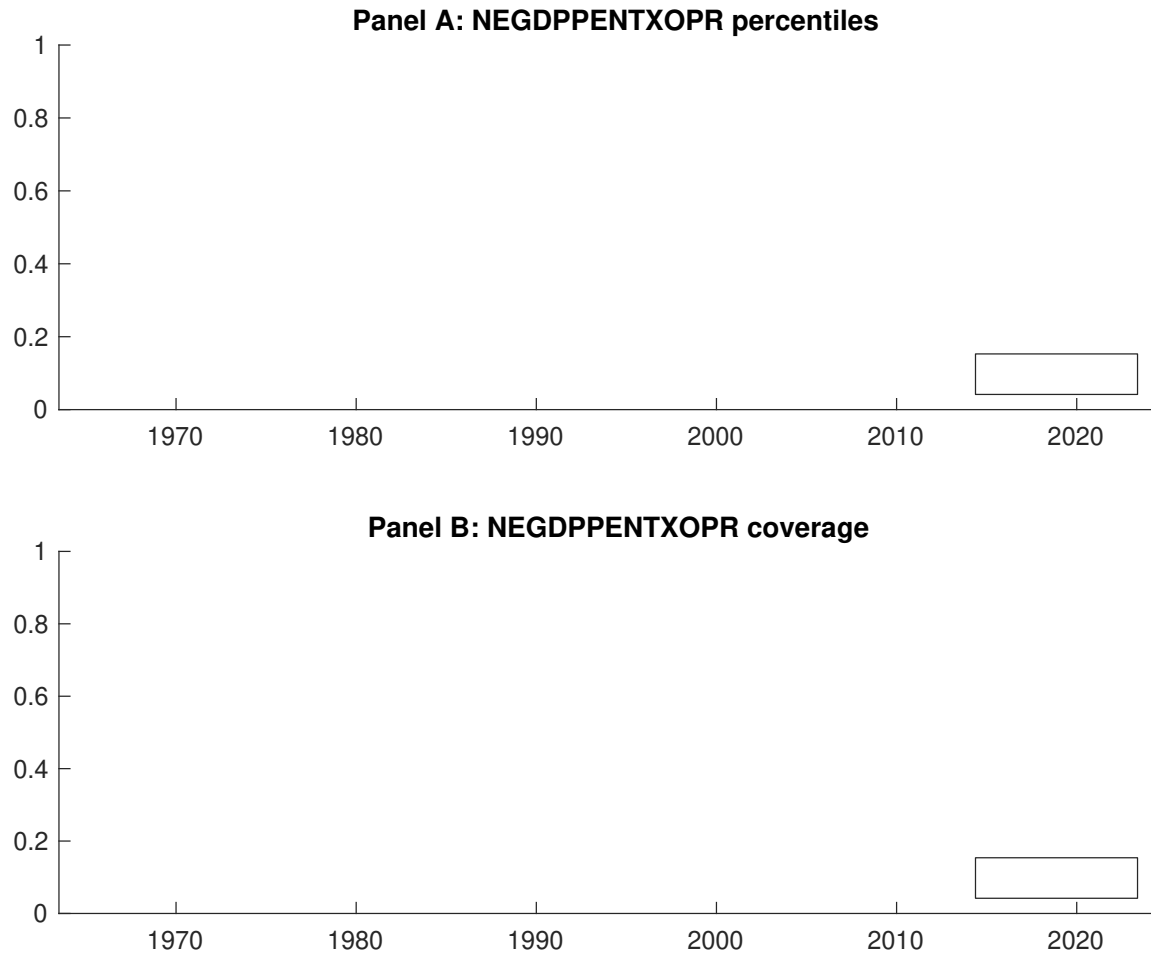


Figure 1: Times series of CEB percentiles and coverage.
This figure plots descriptive statistics for CEB. Panel A shows cross-sectional percentiles of CEB over the sample. Panel B plots the monthly coverage of CEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [3.02]	0.63 [3.81]	0.61 [3.71]	0.65 [3.82]	0.73 [4.10]	0.22 [2.33]
α_{CAPM}	-0.06 [-1.25]	0.06 [1.52]	0.04 [1.03]	0.07 [1.41]	0.14 [2.16]	0.20 [2.11]
α_{FF3}	-0.02 [-0.42]	0.08 [2.28]	0.02 [0.39]	0.01 [0.30]	0.03 [0.62]	0.06 [0.65]
α_{FF4}	0.03 [0.56]	0.05 [1.33]	0.02 [0.42]	0.03 [0.61]	0.01 [0.17]	-0.02 [-0.21]
α_{FF5}	0.09 [2.01]	0.05 [1.46]	-0.06 [-1.42]	-0.06 [-1.39]	-0.06 [-1.15]	-0.16 [-1.98]
α_{FF6}	0.13 [2.66]	0.02 [0.68]	-0.05 [-1.22]	-0.04 [-0.94]	-0.07 [-1.30]	-0.20 [-2.46]
Panel B: Fama and French (2018) 6-factor model loadings for CEB-sorted portfolios						
β_{MKT}	0.94 [82.60]	1.01 [120.52]	1.00 [103.64]	1.01 [90.30]	1.06 [82.84]	0.12 [6.53]
β_{SMB}	-0.12 [-7.29]	-0.11 [-9.07]	0.02 [1.50]	0.10 [6.31]	0.18 [9.57]	0.30 [10.81]
β_{HML}	-0.01 [-0.35]	0.00 [0.11]	0.02 [1.04]	0.06 [2.79]	0.07 [2.77]	0.08 [2.07]
β_{RMW}	-0.20 [-9.22]	0.09 [5.28]	0.16 [8.29]	0.17 [7.61]	0.06 [2.31]	0.26 [7.06]
β_{CMA}	-0.22 [-6.90]	-0.04 [-1.77]	0.11 [3.92]	0.14 [4.33]	0.40 [11.08]	0.63 [11.58]
β_{UMD}	-0.04 [-3.92]	0.04 [4.68]	-0.01 [-1.12]	-0.03 [-2.61]	0.01 [0.99]	0.06 [3.01]
Panel C: Average number of firms (n) and market capitalization (me)						
n	670	639	617	718	820	
me (\$10 ⁶)	2449	2751	2073	1519	1233	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22 [2.33]	0.20 [2.11]	0.06 [0.65]	-0.02 [-0.21]	-0.16 [-1.98]	-0.20 [-2.46]
Quintile	NYSE	EW	0.67 [7.59]	0.66 [7.49]	0.59 [7.64]	0.53 [6.80]	0.55 [7.17]	0.51 [6.57]
Quintile	Name	VW	0.22 [2.18]	0.19 [1.93]	0.04 [0.48]	-0.02 [-0.20]	-0.17 [-2.04]	-0.19 [-2.36]
Quintile	Cap	VW	0.18 [1.96]	0.17 [1.86]	0.05 [0.58]	-0.01 [-0.07]	-0.19 [-2.45]	-0.21 [-2.71]
Decile	NYSE	VW	0.37 [3.00]	0.33 [2.61]	0.20 [1.69]	0.08 [0.66]	-0.03 [-0.28]	-0.11 [-0.97]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [1.92]	0.16 [1.63]	0.03 [0.35]		0.06 [0.70]	0.08 [1.00]
Quintile	NYSE	EW	0.47 [5.07]	0.47 [4.99]	0.39 [4.67]	0.36 [4.35]	0.34 [4.27]	0.33 [4.05]
Quintile	Name	VW	0.17 [1.75]	0.15 [1.47]	0.02 [0.20]		0.06 [0.72]	0.08 [0.93]
Quintile	Cap	VW	0.15 [1.61]	0.14 [1.47]	0.03 [0.35]		0.09 [1.18]	0.11 [1.37]
Decile	NYSE	VW	0.33 [2.63]	0.28 [2.19]	0.17 [1.40]	0.10 [0.82]		

Table 3: Conditional sort on size and CEB

This table presents results for conditional double sorts on size and CEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEB and short stocks with low CEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CEB Quintiles					CEB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.42 [1.74]	0.82 [3.46]	0.90 [3.86]	0.98 [4.16]	0.92 [3.38]	0.50 [4.55]	0.47 [4.26]	0.43 [3.89]	0.34 [3.11]	0.42 [3.90]	0.36 [3.30]
	(2)	0.66 [2.80]	0.71 [3.13]	0.85 [3.98]	0.87 [4.03]	0.92 [4.00]	0.26 [2.74]	0.29 [2.97]	0.20 [2.12]	0.17 [1.72]	0.12 [1.30]	0.10 [1.08]
	(3)	0.47 [2.24]	0.79 [3.82]	0.78 [3.80]	0.84 [4.15]	0.95 [4.53]	0.48 [5.15]	0.48 [5.11]	0.39 [4.30]	0.36 [3.91]	0.31 [3.30]	0.29 [3.09]
	(4)	0.58 [2.98]	0.69 [3.50]	0.76 [3.96]	0.74 [3.92]	0.83 [4.18]	0.25 [2.53]	0.22 [2.20]	0.11 [1.19]	0.12 [1.22]	-0.07 [-0.70]	-0.04 [-0.47]
	(5)	0.52 [3.11]	0.61 [3.54]	0.58 [3.55]	0.51 [3.07]	0.59 [3.62]	0.07 [0.68]	0.07 [0.66]	-0.05 [-0.47]	-0.10 [-1.01]	-0.31 [-3.24]	-0.33 [-3.42]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CEB Quintiles					CEB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	382	383	381	378	371	37	37	32	29	27	
	(2)	108	108	108	108	107	56	56	57	55	55	
	(3)	79	79	79	79	78	97	98	98	98	97	
	(4)	66	67	67	67	66	204	211	214	211	206	
(5)	61	61	61	62	61	1727	1817	1671	1462	1375		

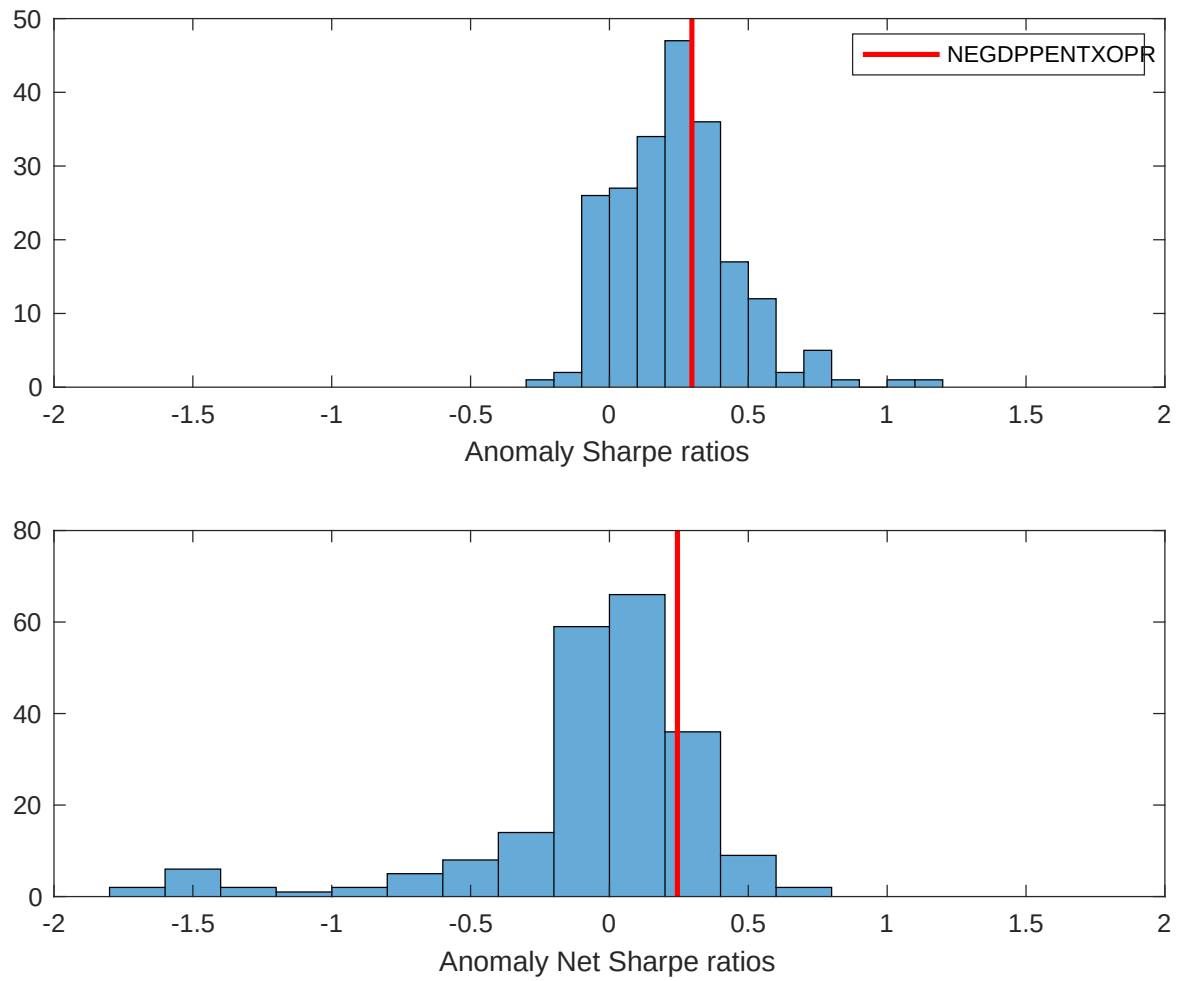


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

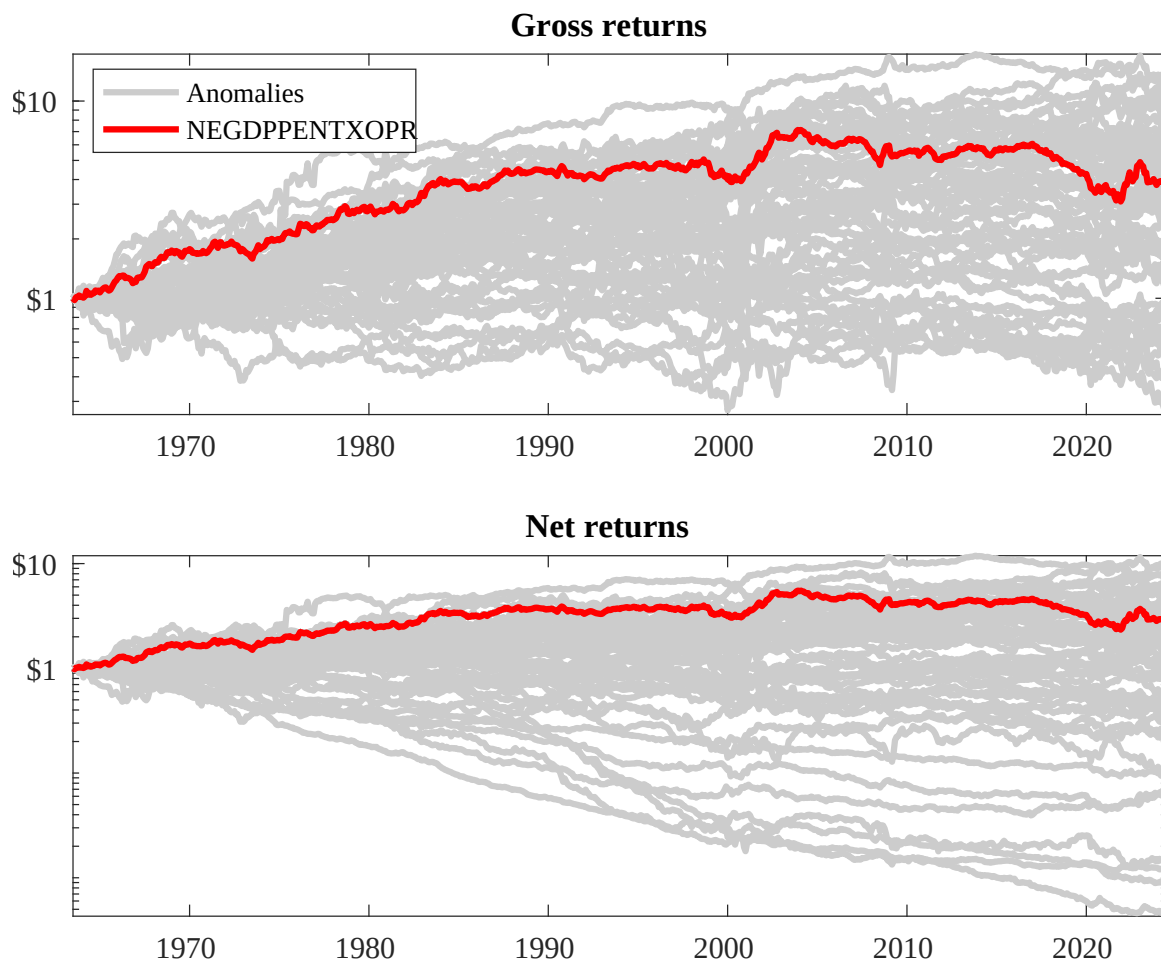
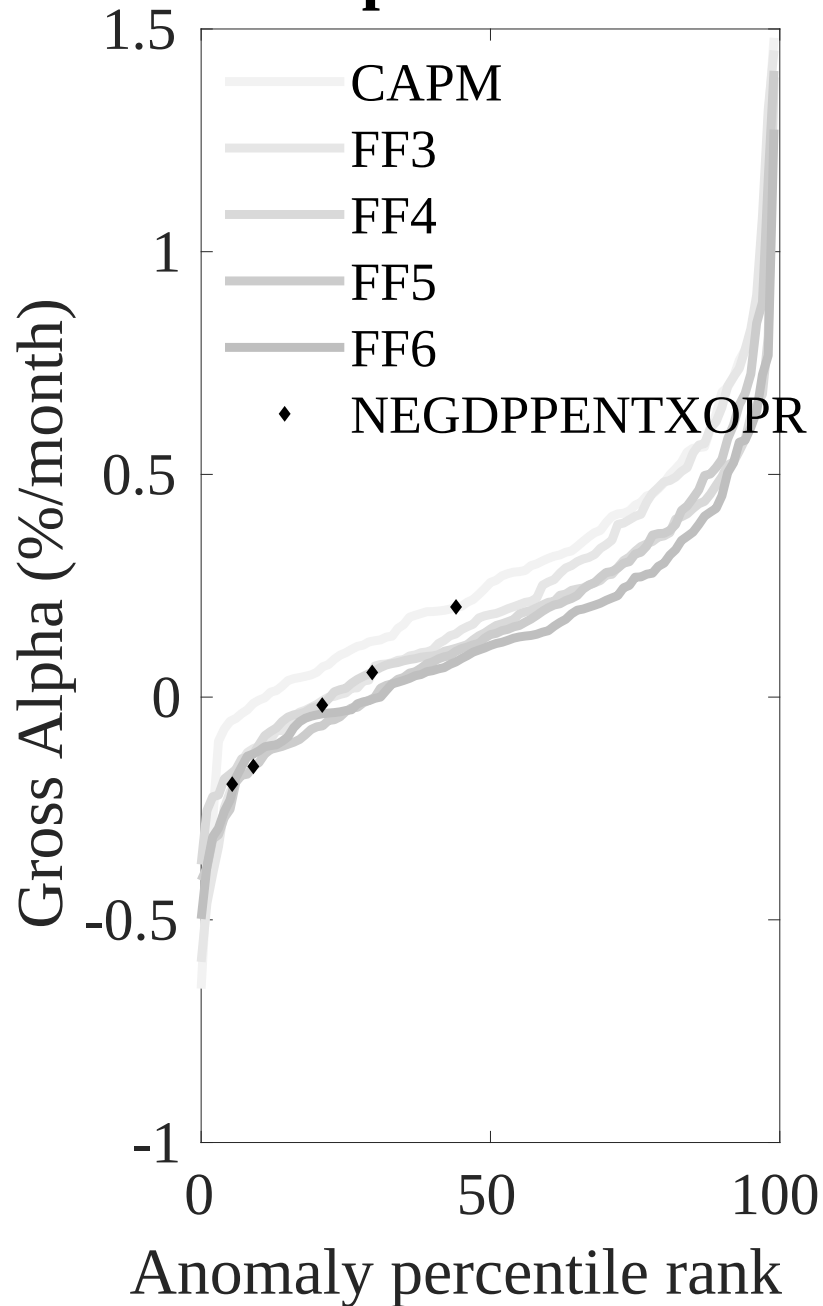


Figure 3: Dollar invested.

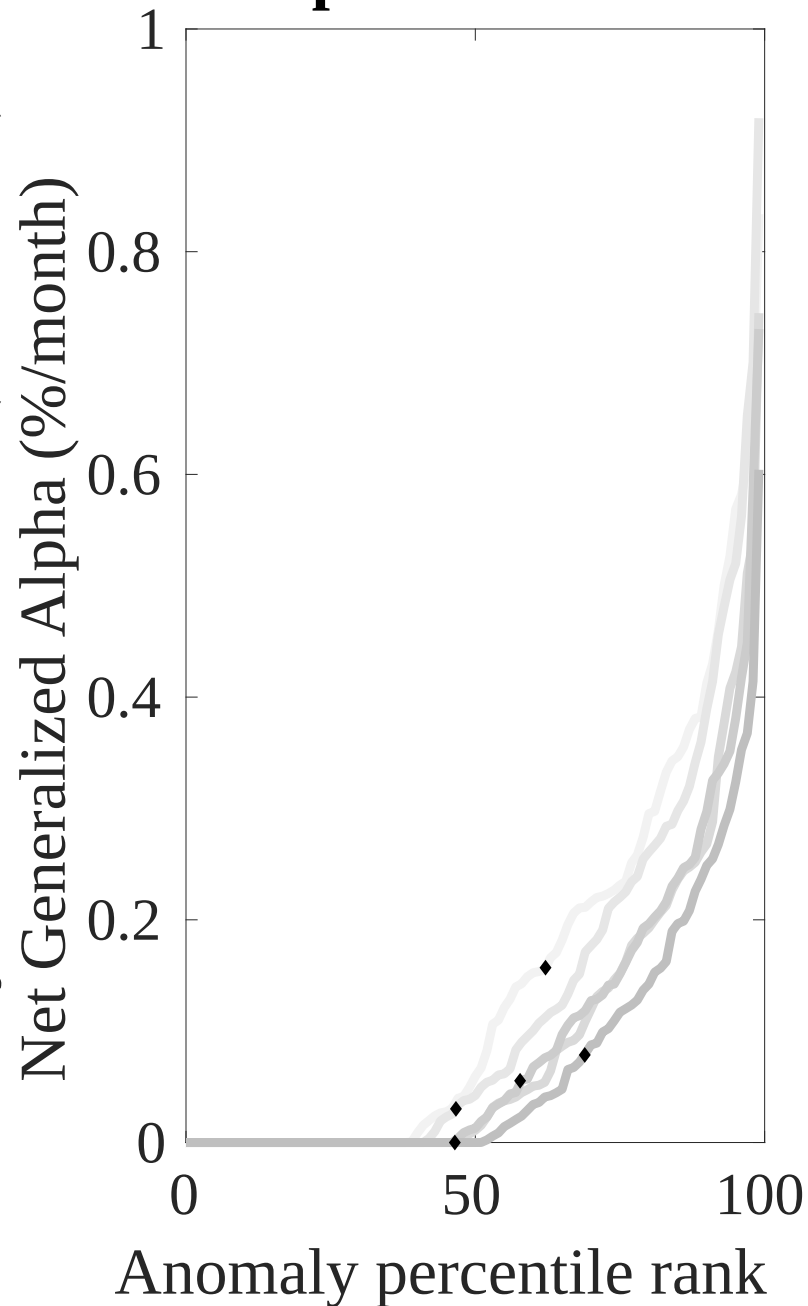
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



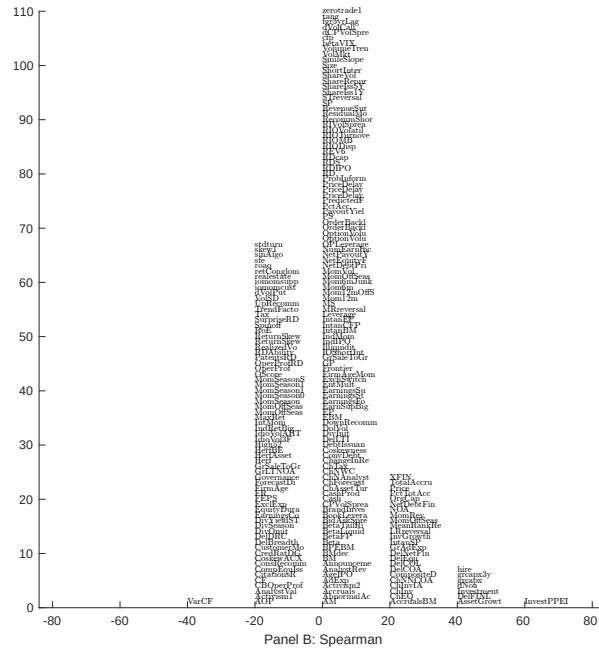
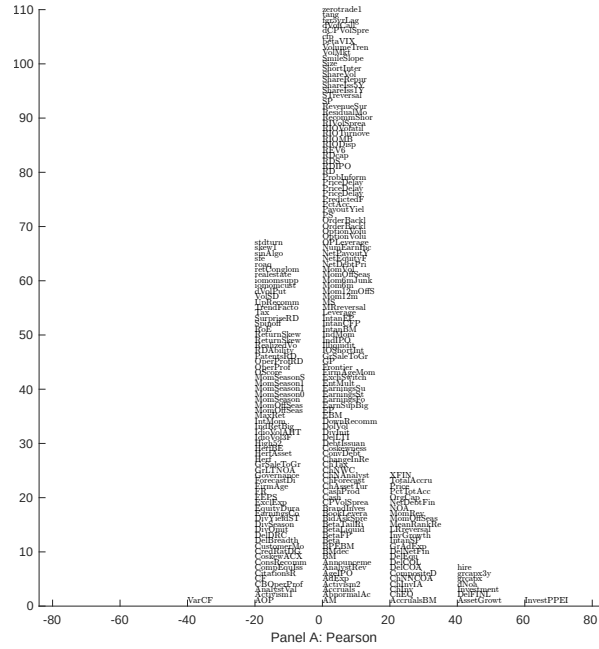


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with CEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

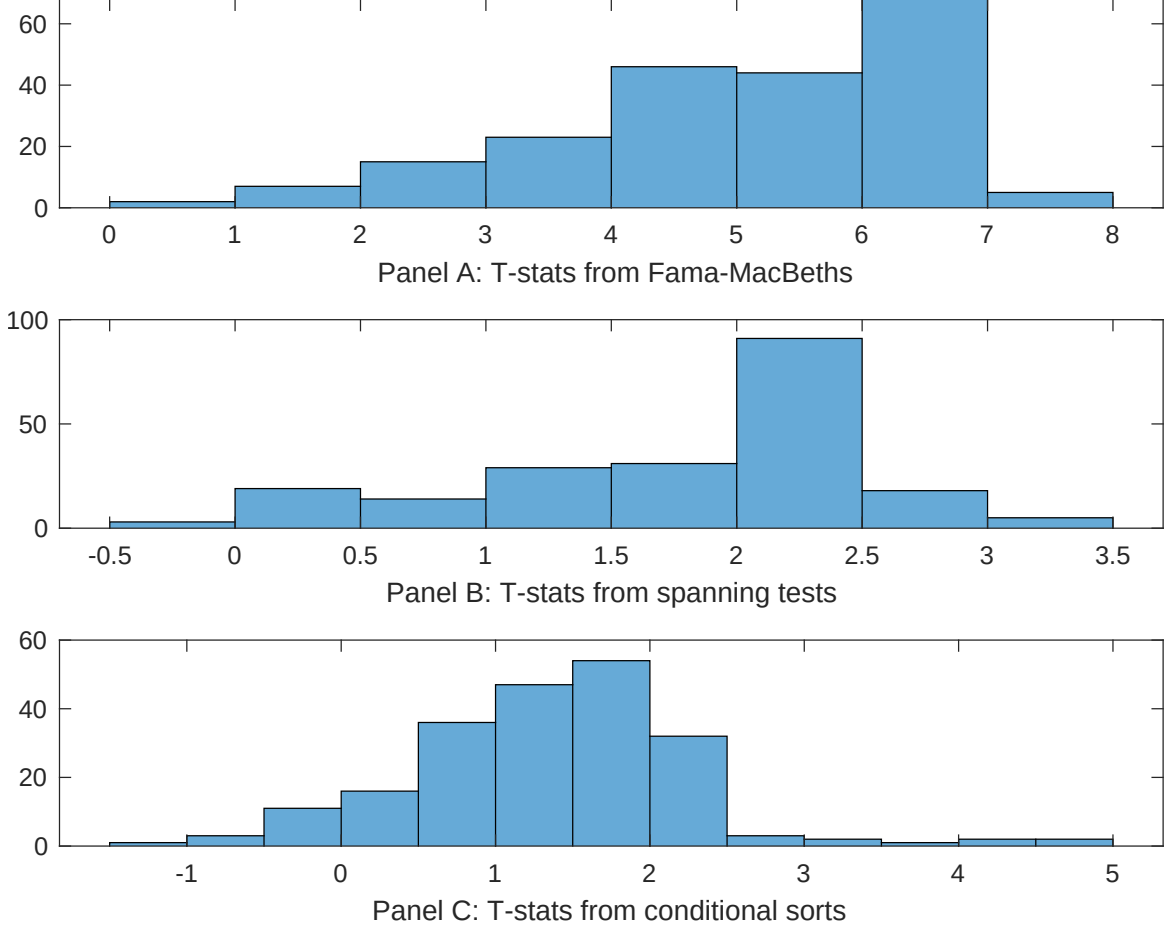


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.14 [6.56]	0.13 [6.01]	0.13 [6.31]	0.13 [6.01]	0.12 [5.66]	0.14 [6.14]
CEB	0.26 [2.72]	0.35 [3.30]	0.41 [4.02]	0.29 [3.15]	0.33 [3.43]	0.45 [5.08]	0.25 [2.24]
Anomaly 1	0.14 [7.51]						-0.16 [-0.71]
Anomaly 2		0.10 [7.25]					0.34 [1.54]
Anomaly 3			0.51 [6.76]				0.14 [1.09]
Anomaly 4				0.91 [7.94]			0.47 [2.83]
Anomaly 5					0.11 [7.16]		0.30 [1.61]
Anomaly 6						0.75 [5.44]	0.15 [1.11]
# months	732	727	727	732	720	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18 [-2.70]	-0.16 [-2.19]	-0.18 [-2.41]	-0.20 [-2.50]	-0.25 [-3.21]	-0.15 [-1.87]	-0.21 [-3.02]
Anomaly 1	50.72 [15.77]						39.87 [11.06]
Anomaly 2		39.69 [10.90]					20.77 [3.47]
Anomaly 3			36.63 [9.92]				4.08 [0.68]
Anomaly 4				21.38 [4.56]			9.79 [2.14]
Anomaly 5					33.58 [7.80]		9.10 [2.06]
Anomaly 6						18.59 [4.29]	-5.82 [-1.45]
mkt	11.68 [7.11]	11.56 [6.53]	11.84 [6.61]	12.33 [6.58]	12.47 [6.82]	12.35 [6.55]	10.99 [6.90]
smb	26.83 [11.29]	21.13 [7.96]	24.44 [9.30]	27.14 [9.91]	29.21 [11.06]	29.80 [10.91]	21.99 [8.95]
hml	0.91 [0.28]	-1.35 [-0.39]	-0.41 [-0.12]	5.78 [1.59]	3.84 [1.08]	2.59 [0.69]	-3.25 [-1.02]
rmw	26.23 [8.19]	23.70 [6.85]	22.98 [6.54]	26.21 [7.16]	26.79 [7.51]	26.96 [7.33]	24.55 [7.86]
cma	24.70 [4.68]	42.71 [7.95]	43.89 [8.05]	37.83 [4.87]	38.58 [6.32]	47.57 [7.23]	6.16 [0.86]
umd	5.36 [3.29]	3.09 [1.75]	2.74 [1.53]	6.12 [3.28]	4.75 [2.62]	4.26 [2.26]	4.08 [2.53]
# months	732	728	728	732	732	713	709
$\bar{R}^2(\%)$	54	46	45	39	42	39	57

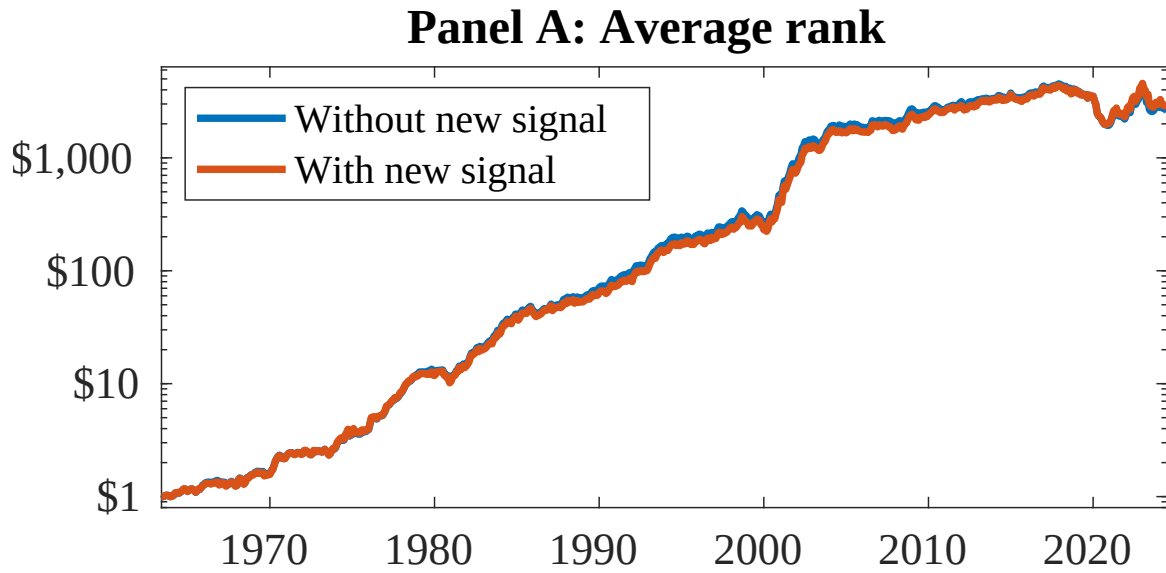


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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